

PDE-Agents: An LLM-Orchestrated Multi-Agent Framework for Automated Finite Element Simulations with Knowledge Graph-Augmented Reasoning

Highlights

- Multi-agent LLM system automates full FEM simulation lifecycle
- GraphRAG knowledge graph provides physics-grounded agent reasoning
- KG Smart integration achieves 100% success on 50 benchmark tasks
- Novel-material experiment proves KG value for unseen properties
- Verified $\mathcal{O}(h^2)$ convergence across all benchmark cases